09-16 第2篇

阅读理解（说明文） Why Do Wild Geese Fly in a "V" Shape?

① In autumn, when cool winds begin to blow across northern fields, a familiar sight appears in the evening sky: **a long line of wild geese** flying south in the shape of a neat letter **"V"**. Country children often run out of their homes to look up and count them. It is beautiful — but it is also one of the **oldest questions** in nature: why do these large birds choose this exact shape for their long journey? Scientists have been studying this for more than fifty years, and today we understand most of the answer.

② The first reason is simple and practical: the V-shape **saves energy**. When a goose flaps its wings, it pushes the air **downwards and outwards**. Just behind and slightly to the side of each wingtip, a small current of rising air forms — a little "air lift" called an **upwash**. The next goose, if it flies in exactly the right place, can ride on this upwash and get free help lifting its body, like a cyclist drafting behind another cyclist. A famous 2014 study on northern bald ibises — close relatives of geese — showed that when a bird flew in the V-position, its heart rate dropped noticeably. Over a long flight, this small daily saving adds up to **many extra kilometres of range**.

③ But energy saving is not the only reason. A V-shape also **helps the birds see each other**. Unlike a tight crowd, where a goose behind may be hidden by the goose in front, a wide V lets almost every bird keep the leader and most team members **in clear sight**. This matters because geese rely on visual signals — a small tilt of a wing, a short call — to stay together. When bad weather suddenly arrives, the whole flock can change direction in seconds, because every bird can see the leader's decision almost at the same moment.

④ A third reason is about **fairness and rest**. The bird at the **very front** of the V has the hardest job: it has no upwash to help it, and it meets the full strength of the wind. Scientists watching long flights have seen the lead goose **fall back** after some time, while another strong goose takes its place at the front. The flock quietly takes turns. A long journey — sometimes more than **3 000 kilometres** between China and Siberia — is only possible because no single bird has to be the leader all day.

⑤ Not every flock flies in a perfect V. On short trips, geese may simply fly in a line, or a loose cluster, or a slight curve called a **"J"-shape**. Scientists have noticed that **younger and less experienced geese** often sit slightly behind the correct spot, losing some of the free lift, simply because they have not learned the trick yet. In some species, the V can also **change sides**, with more birds on the left one day and more on the right the next, depending on the wind. Nature, in other words, prefers "good enough" to "perfect".

⑥ What can we learn from this? A V-shape is **not just a flying habit** — it is a small example of a larger idea: that in a group travelling a long distance, **cooperation saves energy, clear sight keeps everyone together, and taking turns protects the tired**. The next time you look up in October and see a soft, drifting V high above you, remember that those birds are not just flying — they are quietly solving three problems at once, using nothing but wings, eyes, and trust in each other.

注释： upwash 上升气流；drafting 跟随（骑车/飞行时借力）；bald ibis 隐鹮；cluster 群体；J-shape J 字形

根据文章内容，选择最佳答案。

1. What is the main topic of the passage?

A. How geese build their nests.
B. Why wild geese fly in a V-shape and what we can learn from it.
C. How people hunt wild geese in autumn.
D. Why all birds fly south in winter.

2. According to Paragraph 2, how does the V-shape save energy for the geese?

A. Each goose flies directly above the one in front, blocking the wind.
B. Each goose (except the first) rides on a rising "upwash" from the wingtip of the bird in front, which helps lift its body; a study showed the heart rate dropped in the V-position.
C. Geese stop flapping their wings completely when in V-shape.
D. They fly closer to the ground so the air is thicker.

3. Why does the V-shape help the flock "see each other" (Paragraph 3)?

A. Because the shape reflects sunlight more clearly.
B. Because in a wide V almost every bird can keep the leader and most members in clear sight, which helps them change direction quickly together.
C. Because the V-shape makes the geese shine at night.
D. Because the shape scares other birds away.

4. What do scientists say about the leading goose, based on Paragraph 4?

A. The leader stays at the front for the whole journey, no matter how tired.
B. The leader has the easiest job because it flies first.
C. The leader has the hardest job; after some time it falls back and another strong goose takes its place — the flock takes turns.
D. The leader is always the oldest goose in the flock.

5. Which statement best matches the message of the last paragraph?

A. Only smart animals fly south for winter.
B. A V-shape shows three useful ideas at once: cooperation saves energy, clear sight keeps the group together, and taking turns protects the tired.
C. Humans should copy geese and stop using cars.
D. The V-shape is only a beautiful picture with no real use.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】首段提问、末段收束
【解析】全文围绕"大雁为何呈 V 字形飞行"及其启示。A、C、D 与主题不符。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "upwash ... get free help lifting its body ... heart rate dropped noticeably"
【解析】后面的雁利用前方翅尖产生的上升气流省力，研究显示心率下降。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第③段 "a wide V lets almost every bird keep the leader and most team members in clear sight"
【解析】V 形让每只雁都能看到领头与同伴，利于同步转向。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第④段 "The bird at the very front ... has the hardest job ... The lead goose fall back after some time ... The flock quietly takes turns"
【解析】领头任务最重，群体轮流领飞。A、B、D 均错。故选 C。

5.
【答案】B
【题型】主旨/观点题
【定位】第⑥段 "cooperation saves energy, clear sight keeps everyone together, and taking turns protects the tired"
【解析】末段总结三大启示。A、C、D 均错。故选 B。